

UrbanClean

Geospatial Smart City Waste Management & Route Optimization System

UML & System Design Diagram Suite

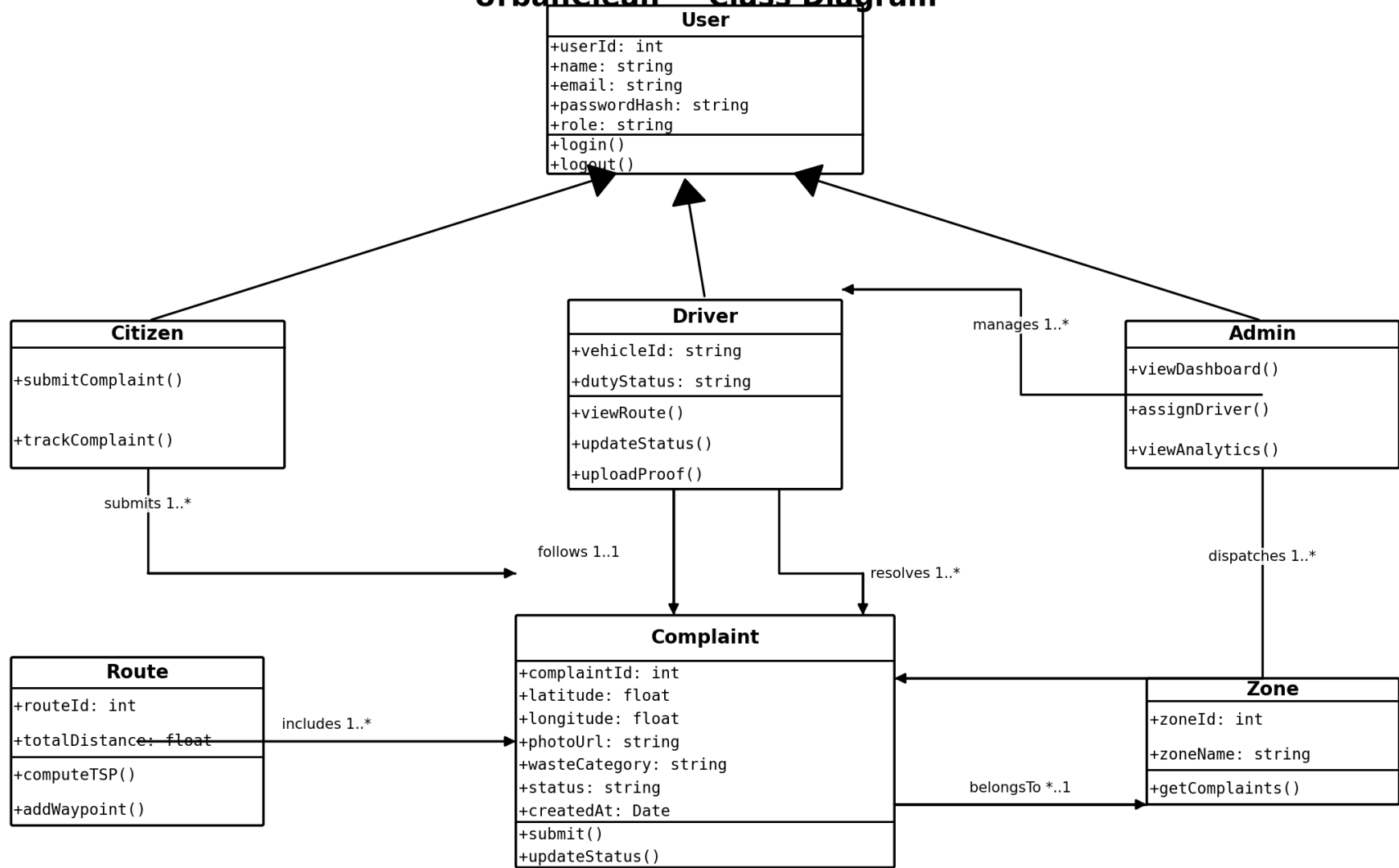
1. Event Table

The table below lists the key system events, what triggers each one, its source, the system's response, and where that response is delivered.

Event	Trigger	Source	Activity	System Response	Destination
Complaint Submission	Citizen clicks Submit with GPS pin + photo	Citizen Portal	Create Ticket (Validate & Store)	Creates ticket (status = Pending), notifies Admin Console	Admin Console
User Authentication	User submits login credentials	Citizen / Driver / Admin	Authenticate & Create Session	Validates salted hash, creates session, redirects to role dashboard	Respective Dashboard
Task Assignment	New complaint enters dispatch queue	Admin / Dispatch System	Auto/Manual Driver Assignment	Maps ticket to the nearest on-duty driver covering that ward	Driver App
Route Optimization	New waypoint added to a driver's schedule	OSRM Routing Engine	Compute TSP Optimized Route	Recomputes TSP-based shortest, fuel-efficient path and updates the map	Driver Map (Leaflet)
Duty Status Toggle	Driver marks On-Duty / Off-Duty	Driver Portal	Update Driver Availability	Updates driver roster and availability count	Admin Analytics Dashboard
Complaint Resolution	Driver uploads proof-of-cleanup photo	Driver Portal	Verify Proof & Close Ticket	Marks ticket status Completed (green), logs resolution timestamp	Citizen Tracker + Admin Console

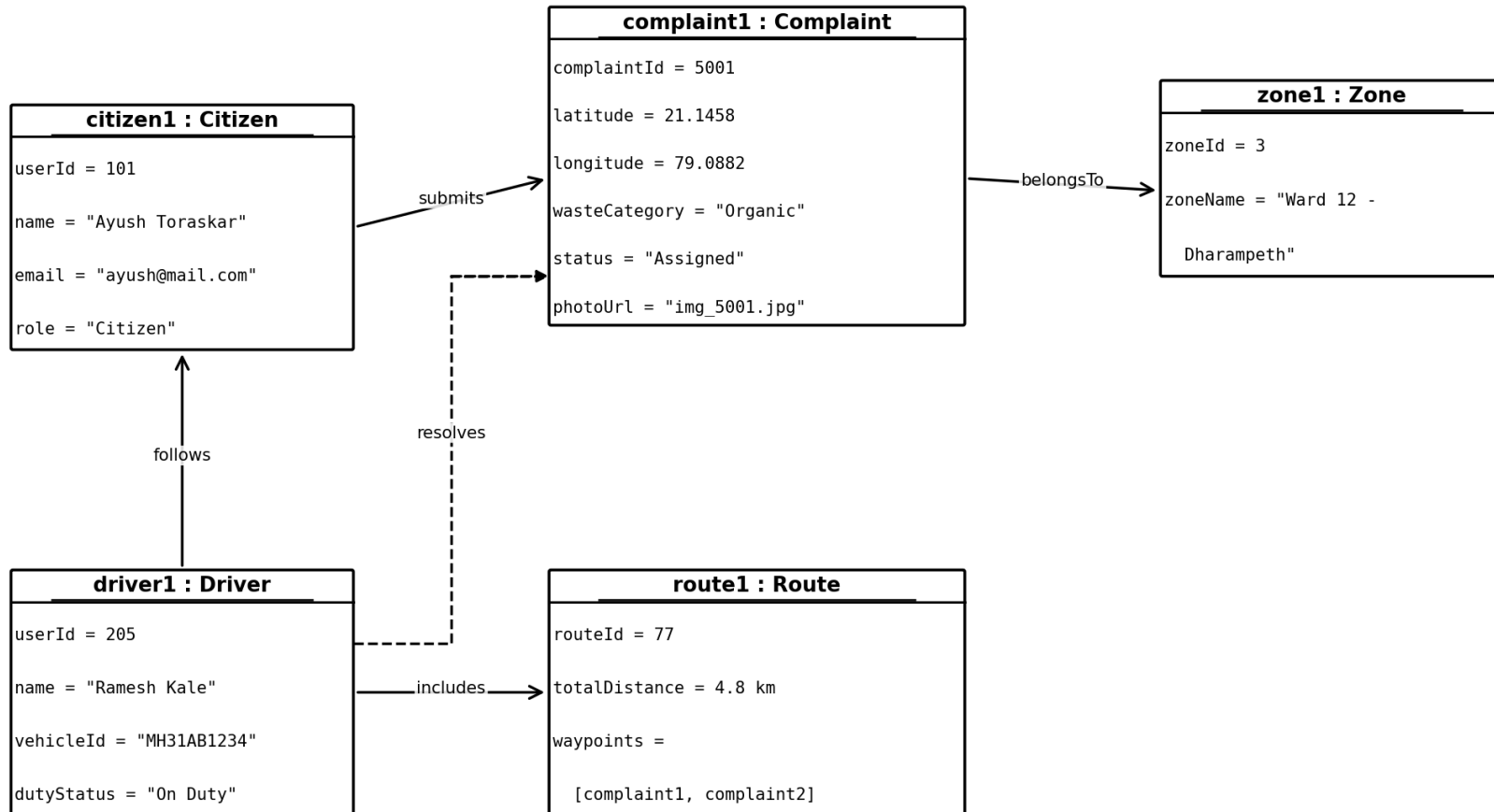
2. Class Diagram

UrbanClean — Class Diagram

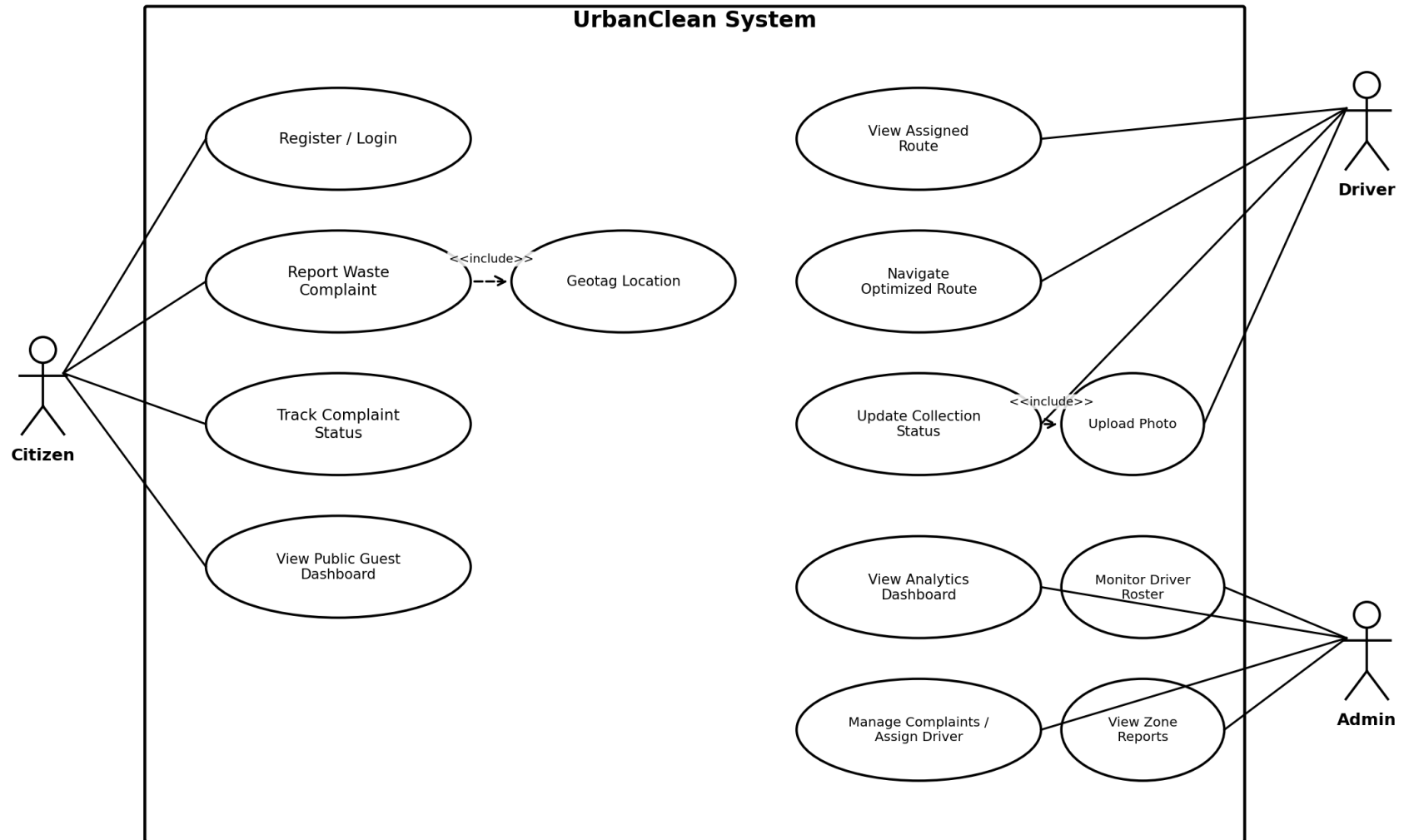


3. Object Diagram

UrbanClean — Object Diagram (Runtime Snapshot)

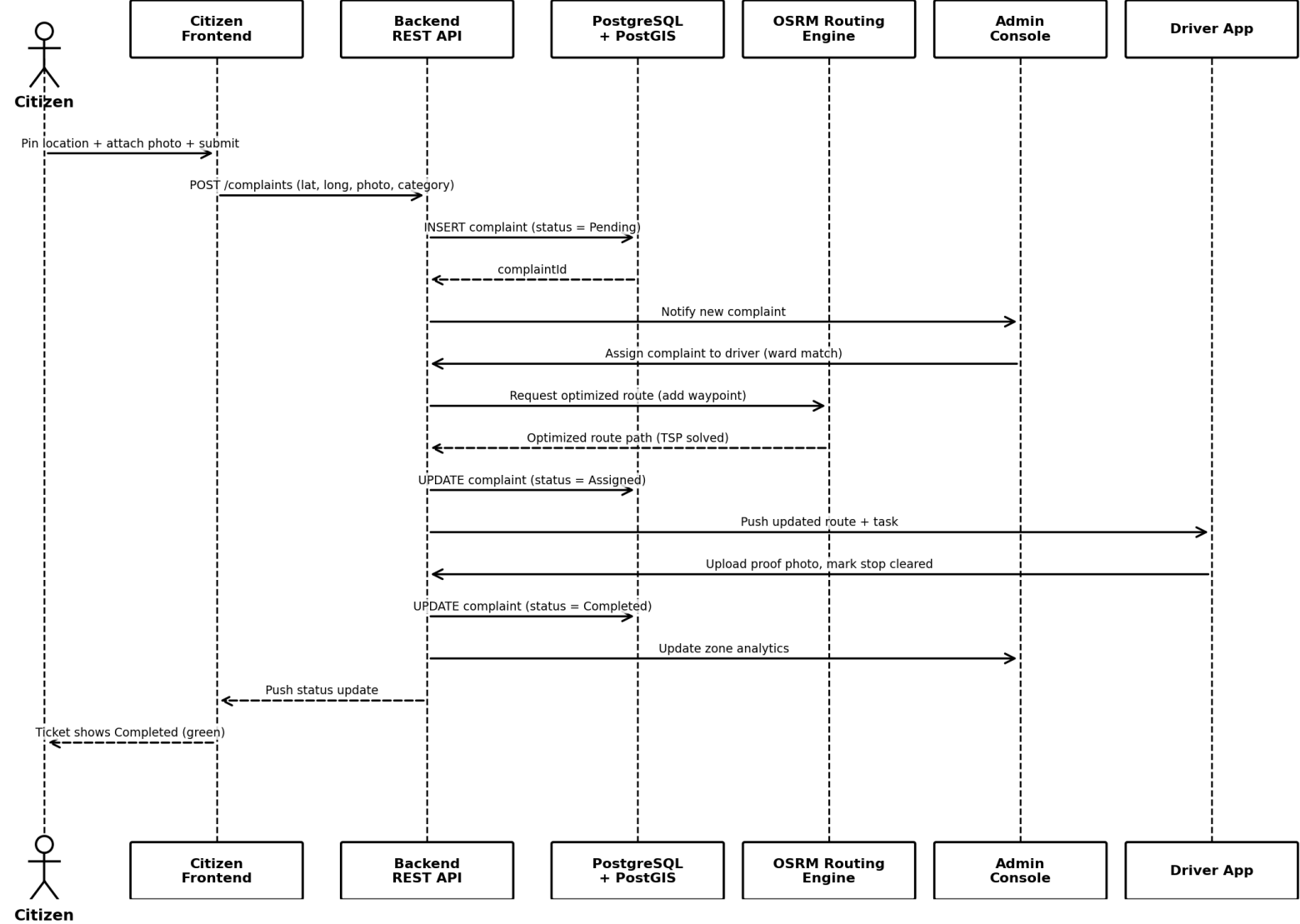


4. Use Case Diagram



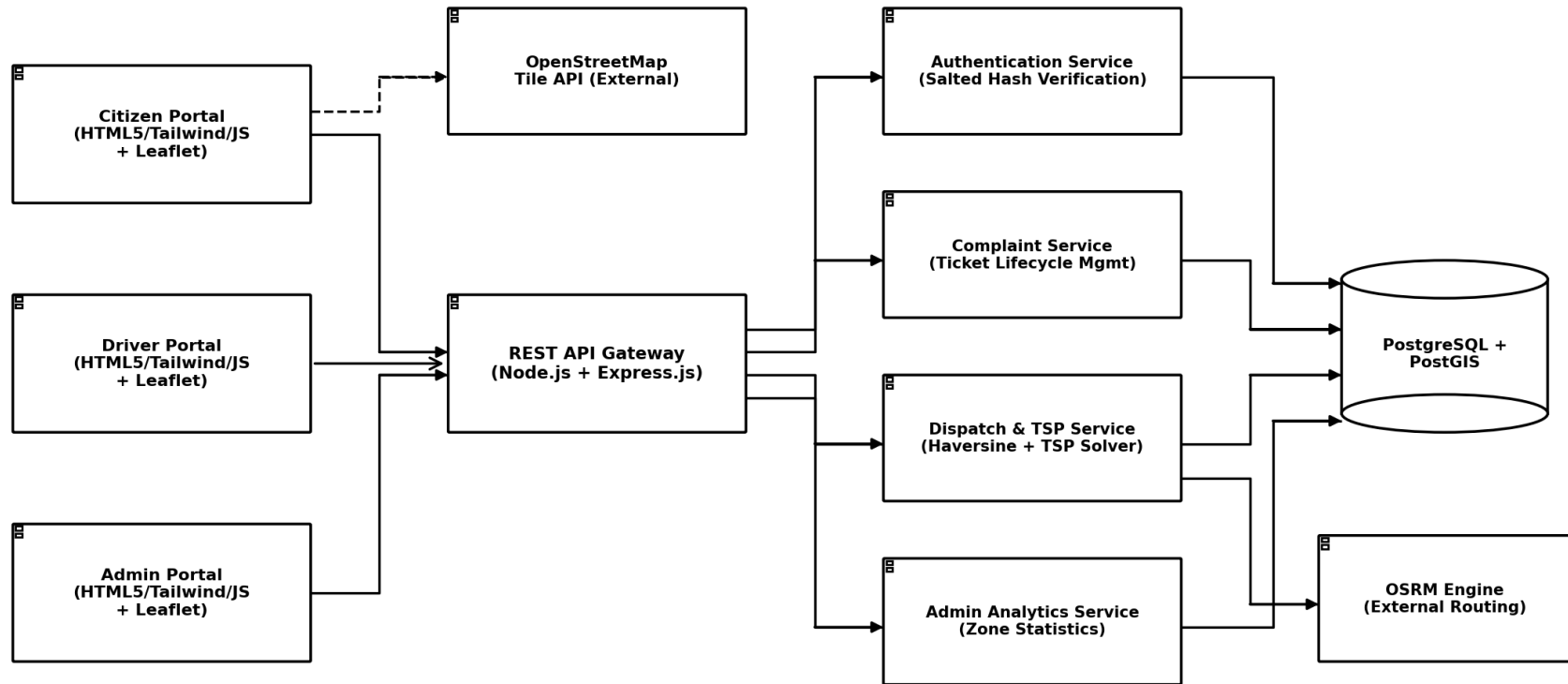
5. Sequence Diagram

UrbanClean — Sequence Diagram: Complaint Lifecycle



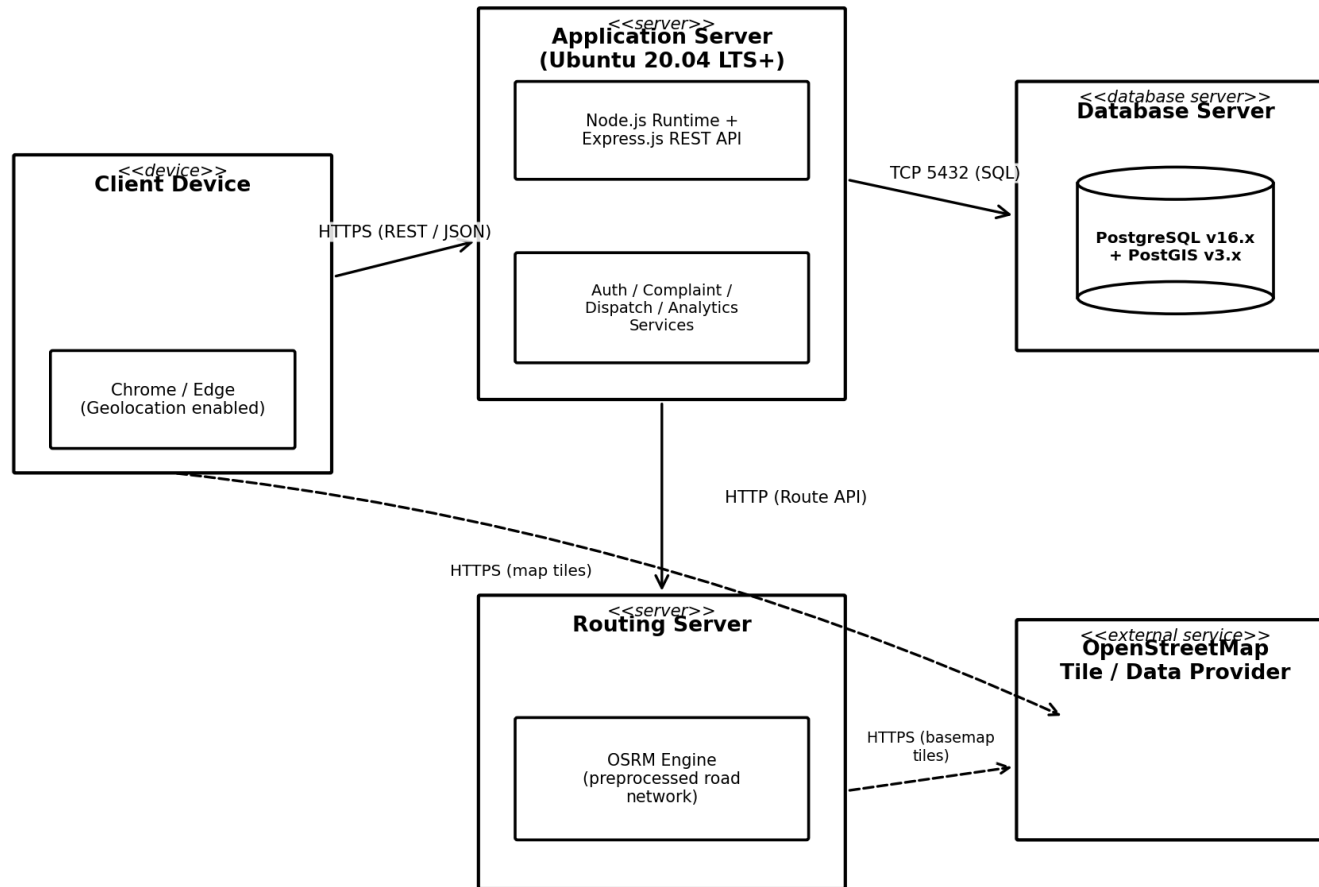
6. Component Diagram

UrbanClean — Component Diagram



7. Deployment Diagram

UrbanClean — Deployment Diagram



8. Activity Diagram

UrbanClean — Activity Diagram

(Complaint Validation & Dispatch Routing)

